Haoyu Wang

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https://carlwhy-28.github.io

Education

University of Pittsburgh, School of Medicine — **Pittsburgh, PA**
Ph.D. in Biomedical Informatics — *Aug 2025 - 2029*

- Focus: Computational Biology, Machine Learning, Spatial Transcriptomics

Beijing Forestry University — **Beijing, China**
B.E. in Computer Science and Technology — *Sep 2020 - Jun 2024*

Professional Experience

University of Pittsburgh Medical Center — **Pittsburgh, PA**
Computational Biology Intern — *Aug 2024 – Jul 2025*

- Designed and implemented dual-graph attention neural networks (DGAT) for spatial multi-omics data integration, enabling accurate protein expression imputation from transcriptomic and spatial features
- Performed downstream pathway and transcription factor activity inference using statistical modeling and enrichment analysis to identify key regulatory signals in tumor microenvironments

Publications

Wang, H., Batz, C., Osmanbeyoglu, H.U., *DGAT: A Dual-Graph Attention Network for Inferring Spatial Protein Landscapes from Transcriptomics.bioRxiv* [Preprint], 2025 Jul 9. doi: 10.1101/2025.07.05.662121
Selected for oral presentation at ISMB 2025, workshop at MLCB 2025. *Under revision at Nature Communications*

Zhang, L.; Sagan, A.; Qin, B.; **Wang, H.**; Kim, E.; Hu, B.; Osmanbeyoglu, H.U., *STAN, a computational framework for inferring spatially informed transcription factor activity.bioRxiv* [Preprint], 2025 Jul 9. doi: 10.1101/2024.06.26.600782
Accepted by Nucleic Acids Research

Research Experience

DGAT: Spatial Protein Landscapes Imputation from Transcriptomics — **Pittsburgh, PA**
UPMC Research Project — *Sep 2024 – May 2025*

- Designed and developed a **spatially-awared GAT-based model for protein imputation** from mRNA and spatial data
- Improved protein-wise correlation by 5%–20% over 7 prior methods in sample-hold-out benchmarking
- In spatial transcriptomic data lacking protein level but with pathological annotations, achieved 1%–3% Adjusted Rand Index gains in spatial domain clustering

STAN: a computational framework inferring transcription factor activity — **Pittsburgh, PA**
UPMC Research Project — *Aug 2024 – Dec 2024*

- Performed downstream analysis of pancreatic cancer Spatial Transcriptomics data, utilizing statistical models to infer transcription factor and pathway activities across patient groups
- Applied unsupervised deep learning methods to **whole slide image feature extraction** for downstream modeling
- Identification and visualization of significant patterns to uncover biological insights related to cancer progression

Technical Skills

- **Programming Languages:** Proficient – Python, SQL; Intermediate – Java, C++, MATLAB, R; Basic
- **Machine Learning/AI:** PyTorch, TensorFlow, scikit-learn, GNNs, Transformer, Computer Vision
- **Bioinformatics:** Spatial Transcriptomics, Single-cell Analysis, Genomics, Pathway Modeling
- **Tools & Platforms:** Linux (Ubuntu), Git, Jupyter, SLURM, VSCode